Renal clearance (sample lecture)

Synthetic demo lecture for med-student-ai-demo

Learning objectives: (1) define clearance; (2) estimate GFR; (3) interpret clearance relative to inulin

Definition of clearance

Clearance = (U x V) / P: the plasma volume cleared of a substance per unit time

U = urine concentration, V = urine flow rate, P = plasma concentration

Units are mL/min

Inulin and GFR

Inulin is freely filtered and neither reabsorbed nor secreted

So inulin clearance equals GFR (about 125 mL/min in a healthy adult)

Creatinine clearance approximates GFR but slightly overestimates it (small tubular secretion)

Interpreting clearance

Clearance below GFR: net reabsorption (for example glucose, which is 0 at normal plasma levels)

Clearance above GFR: net secretion (for example para-aminohippuric acid, PAH)

PAH clearance approximates renal plasma flow (about 600 mL/min)